EARTH SCIENCE

Earth Science

The Earth Science major is a program leading to the Bachelor of Science (B.S.) or Bachelor of Arts (B.A.).

The B.S. program emphasizes an interdisciplinary approach to Earth system processes at multiple temporal and spatial scales. The degree is suitable for students interested in identifying and monitoring interactions between the lithosphere, hydrosphere, atmosphere, and human activities. It prepares students for consulting and geotechnical careers in the private and government sectors.

The B.A. emphasizes an interdisciplinary field and, in combination with a secondary major in education, prepares the student primarily for a career in secondary education.

Student Learning Outcomes

Upon graduation, students should be able to:

- Demonstrate the ability to think critically;
- Exhibit a broad understanding of Earth systems and processes;
- Demonstrate scientific literacy and its application to scientific inquiry and societal concerns;
- Demonstrate proficiency in data collection, management, and analysis including understanding sources of error and/or uncertainty;
- Demonstrate competency with geoscience-specific techniques and field methods.
- Read and critically evaluate relevant literature and information;
- Use appropriate tools from chemistry, physics, biology, mathematics, and data science to solve discipline-specific problems;
- Present information effectively in written and oral forms;
- Work in a team environment in alignment with the ISU principles of community;
- Work independently;
- Attain employment in the geosciences or related fields or pursue graduate studies.

Students can pursue either the Bachelor of Science (B.S.) or the Bachelor of Arts (B.A.) degree with a major in Earth Science.

EARTH SCIENCE, B.S.

Required courses for the B.S. include:

GEOL 1000	How the Earth Works	3
or GEOL 1010	Environmental Geology: Earth in Crisis	
or GEOL 2010	Geology for Engineers and Environmental Scientists	
GEOL 1000L	How the Earth Works: Laboratory	1
GEOL 3020	Summer Field Studies	6
GEOL 3150	Mineralogy and Earth Materials	3
GEOL 3150L	Laboratory in Mineralogy and Earth Materials	1
GEOL 3560	Global Tectonics	3
GEOL 3570	Geological Mapping and Field Methods	1
GEOL 3680	Sediments and Depositional Environments	3
GEOL 4020	Watershed Hydrology	3
GEOL 4110	Hydrogeology	4
GEOL 4520	Intro GIS for Geoscientists	3
GEOL 4790	Surficial Processes	3
And 9 credits of Earth Science electives in agronomy, environmental science, meteorology, or other approved areas.		9
Total Credits		**43**

Required supporting courses for the B.S. include:

CHEM 1770	General Chemistry I	4
CHEM 1770L	Laboratory in General Chemistry I	1
CHEM 1780	General Chemistry II	3
CHEM 1780L	Laboratory in College Chemistry II	1
MATH 1600	Survey of Calculus	4
PHYS 1310	General Physics I	4
PHYS 1310L	General Physics I Laboratory	1
PHYS 1320	General Physics II	4
PHYS 1320L	General Physics II Laboratory	1
STAT 1040	Introduction to Statistics	3
STAT 3201	Intermediate Statistical Concepts and Methods	4
Total Credits		**30**

Communication Proficiency requirement: According to the university-wide Communication Proficiency Grade Requirement (https://catalog.iastate.edu/academics/#communicationproficiencypolicytext), students must demonstrate their communication proficiency by earning a grade of C or better in ENGL 2500. The department requires a grade of C or better in the below communication courses.

ENGL 1500	Critical Thinking and Communication	3
ENGL 2500	Written, Oral, Visual, and Electronic Composition	3
or ENGL 2500H	Written, Oral, Visual, and Electronic Composition: Honors	
ENGL 3090	Proposal and Report Writing	3
or ENGL 3140	Technical Communication	
or ENGL 3020	Business Communication	
or JLMC 3470	Science Communication	
Total Credits		**9**

Earth Science Undergraduate Majors (B.A. or B.S.) may not earn a Geology Undergraduate Minor.

As majors in the College of Liberal Arts and Sciences, Earth Science students must meet College of Liberal Arts and Sciences (https://catalog.iastate.edu/collegeofliberalartsandsciences/#lascollegerequirementstext) and University-wide requirements (https://catalog.iastate.edu/collegescurricula/) for graduation in addition to those stated above for the major.

Students in all ISU majors must complete a three-credit course in U.S. Cultures and Communities and a three-credit course in International Perspectives. Check (http://www.registrar.iastate.edu/courses/div-ip-guide.html) for a list of approved courses.

LAS majors require a minimum of 120 credits, including a minimum of 45 credits at the 3000/4000 level in addition to the LAS world language and cultures and career preparation requirement (LAS 2030 Professional Career Preparation). At least 8 credits in the major from 3000+ courses must earn grade C or better. The average grade of all courses in the major must be 2.0 or higher.

Earth Science, B.A.

Earth Science, B.A. students are required to fulfill either the Earth Science Teacher Preparation Focus along with the Secondary Major in Education or another secondary area of expertise (Minor, Certificate, Additional Major, Secondary Major). Suggested secondary areas of expertise include Geographic Information Systems Undergraduate Minor, Environmental Studies Secondary Major, Sustainability Undergraduate Minor, Science Communication Certificate; this list provides examples but is not comprehensive. Earth Science Undergraduate Majors (B.A. or B.S.) may not earn a Geology Undergraduate Minor.

Required courses for the B.A. include:

GEOL 1120	Geoscience Orientation	1
GEOL 1000	How the Earth Works	3
or GEOL 1010	Environmental Geology: Earth in Crisis	
or GEOL 2010	Geology for Engineers and Environmental Scientists	
GEOL 1000L	How the Earth Works: Laboratory	1
GEOL 2020	History of the Earth	3
GEOL 2020L	History of the Earth: Laboratory	1
MTEOR 2060	Introduction to Weather and Climate	3
ASTRO 1200	The Sky and the Solar System	3
or ASTRO 1500	Stars, Galaxies, and Cosmology	
ENSCI 2500	Environmental Geography	3
Climate/Human Impacts (Choose one course)		
ENSCI 3180	Introduction to Ecosystems	3
or MTEOR 4040	Global Change	
or GEOL 3240	Energy and the Environment	
or GEOL 4110	Hydrogeology	
Astronomy (Choose one course)		
ASTRO 1200	The Sky and the Solar System	3
or ASTRO 1500	Stars, Galaxies, and Cosmology	
Geology (Choose two courses, minimum 6 credits)		6-8
Any 3000- or 4000-level GEOL courses not used above		
Total Credits		**30-32**

Required supporting courses for the B.A. include:

Choose one Chemistry and Lab combination:		5
CHEM 1630 & 1630L	College Chemistry and Laboratory in College Chemistry	
CHEM 1670 & 1670L	General Chemistry for Engineering Students and Laboratory in General Chemistry for Engineering	
CHEM 1770 & 1770L	General Chemistry I and Laboratory in General Chemistry I	
PHYS 1310 & 1310L	General Physics I and General Physics I Laboratory	5
MATH 1510	Calculus for Business and Social Sciences	3
or MATH 1600	Survey of Calculus	
or MATH 1650	Calculus I	
Total Credits		**13**

Communication Proficiency requirement: According to the university-wide Communication Proficiency Grade Requirement (https://catalog.iastate.edu/academics/#communicationproficiencypolicytext), students must demonstrate their communication proficiency by earning a grade of C or better in ENGL 2500. The department requires a grade of C or better in ENGL 3090 or ENGL 3140.

ENGL 1500	Critical Thinking and Communication	3
ENGL 2500	Written, Oral, Visual, and Electronic Composition	3
or ENGL 2500H	Written, Oral, Visual, and Electronic Composition: Honors	
One of the following:		
ENGL 3090	Proposal and Report Writing	3
or ENGL 3140	Technical Communication	
Total Credits		**9**

As majors in the College of Liberal Arts and Sciences, Earth Science students must meet College of Liberal Arts and Sciences (https://catalog.iastate.edu/collegeofliberalartsandsciences/#lascollegerequirementstext) and University-wide requirements (https://

catalog.iastate.edu/collegescurricula/) for graduation in addition to those stated above for the major.

Students in all ISU majors must complete a three-credit course in U.S. Cultures and Communities and a three-credit course in International Perspectives. Check (http://www.registrar.iastate.edu/courses/div-ip-guide.html) for a list of approved courses.

LAS majors require a minimum of 120 credits, including a minimum of 45 credits at the 3000/4000 level in addition to the LAS world language and cultures and career preparation requirement (LAS 2030 Professional Career Preparation). At least 8 credits in the major from 3000+ courses must earn grade C or better. The average grade of all courses in the major must be 2.0 or higher.

Teacher Preparation Focus

Earth Science majors seeking certification to teach Earth Science in secondary schools follow the requirements of the B.A. degree in Earth Science. In addition, they must take the complementary coursework listed below and meet all of the requirements of the Secondary Major in Education. (https://catalog.iastate.edu/collegeofhumansciences/educationsecondary/#curriculumtext) Some of these courses, and some of the required coursework for the Secondary Major in Education (https://catalog.iastate.edu/collegeofhumansciences/educationsecondary/#curriculumtext), may also apply to LAS general education requirements.

Complementary Coursework		
PSYCH 2300	Developmental Psychology	3
EDUC 4180	Secondary Science Methods I	3
EDUC 4190	Secondary Science Methods II	3
One course in American History or Government		

Notes: Teacher license requirements are established by the Iowa Department of Education and the Iowa Board of Educational Examiners and are subject to change. Recent changes may not be reflected in this catalog, but advisers and faculty will be aware. Some students pursuing the Earth Science decide to complete the Earth Science major and continue their studies as graduate students in Iowa State's Science Education, Masters of Teaching (M.A.T.) program (https://education.iastate.edu/graduate-students/graduate-programs/division-of-teaching-learning-leadership-and-policy/educator-prep-programs-for-licensure-recommendation/science-education-m-a-t/endorsements/).

EARTH SCIENCE, B.S.

Below is a suggested pathway for new majors. This plan is an example only; students should discuss their graduation plan with their advisor.

Freshman

Fall	Credits	Spring	Credits
ENGL 1500	3	CHEM 1780	3
GEOL 1000 or 1010	3	CHEM 1780L	1
GEOL 1000L	1	STAT 1040	3
CHEM 1770	4	Arts-and-Humanities Choice	3
CHEM 1770L	1	Elective	3-4
LIB 1600	1		
MATH 1600	4		
	17		**13-14**

Sophomore

Fall	Credits	Spring	Credits
ENGL 2500	3	Arts-and-Humanities Choice	3
GEOL 3150	3	PHYS 1320	4
GEOL 3150L	1	PHYS 1320L	1
PHYS 1310	4	World Language	3
PHYS 1310L	1	STAT 3201	4
LAS 2030	1		
World Language	3-4		
	16-17		**15**

Junior

Fall	Credits	Spring	Credits	Summer	Credits
GEOL 3680	3	GEOL 4520	3	GEOL 3020	6
GEOL 3570	1	Social-Science Choice	3		
Elective	6-7	GEOL 3560	3		
Arts-and-Humanities Choice	3	Elective	3-4		
	13-14		**12-13**		**6**

Senior

Fall	Credits	Spring	Credits
GEOL 4020	3	Electives	3
Social Science Choice	3	Arts-and-Humanities Choice	3

ENGL 3090, 3020, 3140, or JLMC 3470	3	Social-Science Choice	3
GEOL 4790	3	GEOL 4110	4
Elective	3		
	15		**13**

Total Credits: 120-124

Earth Science, B.A.

Below is a suggested pathway for the B.A. major in Earth Science with a secondary major in Education. This plan is an example only; students should discuss their graduation plan with their advisor.

Freshman			
Fall	**Credits**	**Spring**	**Credits**
ENGL 1500	3	EDUC 2040	3
LIB 1600	1	GEOL 2020	3
GEOL 1000	3	GEOL 2020L	1
GEOL 1000L	1	MTEOR 2060	3
CHEM 1770, 1630, or 1670	4	MATH 1510	3
CHEM 1770L, 1630L, or 1670L	1	PSYCH 2300	3
GEOL 1120	1		
EDUC 2190	1		
	15		**16**
Sophomore			
Fall	**Credits**	**Spring**	**Credits**
ENGL 2500	3	PSYCH 3330 (social science)	3
PHYS 1310	4	Arts and Humanities Option [1]	3
PHYS 1310L	1	Climate/Human impacts choice	3
LAS 2030	1	Geology choice (3000-4000 level)	3
EDUC 2020	3	Astronomy choice	3
EDUC 2800L	0.5		
ENSCI 2500	3		
Apply/Accepted to Educator Preparation Program			
	15.5		**15**
Junior			
Fall	**Credits**	**Spring**	**Credits**
EDUC 4180	3	EDUC 4190	3
EDUC 3950	3	ENGL 3090 or 3140	3
EDUC 3800J	1-2	EDUC 4800J	2
HIST 2800 or 2810 (humanities)	3	COMST 2110, SPCM 2120, or THTRE 3580	3
ASTRO 1200	3	World Language	3-4
Geology choice (3000-4000 level)	3	Arts and Humanities Option[1]	3
		Apply for student teaching	
	16-17		**17-18**
Senior			
Fall	**Credits**	**Spring**	**Credits**
SPED 4010	3	EDUC 4170J	8
EDUC 4060	3	EDUC 4170J	8
World Language	3-4		
Arts and Humanities Option	3		
Social Science Option[1]	3		
Apply for graduation			
	15-16		**16**

Total Credits: 125.5-128.5

Students must take an American History (counts as humanities) or American Government (counts as social science).

1 Choose from list of approved courses available from an advisor.

Graduate Programs

The department offers programs leading to the Master of Science and Doctor of Philosophy with majors in Geology, Earth Science, and Meteorology. Students desiring a major in the above fields normally will have a strong undergraduate background in the physical and mathematical sciences. Individuals desiring to enter a graduate program are evaluated by considering their undergraduate preparation and performance along with their expressed goals in the statement of purpose. All prospective students should reach out to individual faculty members who they wish to work with prior to applying.

Programs of study are designed on an individual basis in accordance with requirements of the Graduate College and established requirements for each departmental major. Additional coursework is normally taken in complementary areas such as aerospace engineering, agronomy (soil science), chemistry, civil and construction engineering, computer engineering, computer science, engineering mechanics, environmental science, materials engineering, mathematics, mechanical engineering, microbiology, physics, or statistics. Departmental requirements provide a strong, broad background in the major and allow considerable flexibility in the program of each individual.

A dissertation is required of all Ph.D. candidates.

M.S. students in Geology are required to complete a thesis. The M.S. in Earth Science is available to students electing the non-thesis (Creative Component) option in Geology or Meteorology.

Graduates in Geology specialize in a sub-discipline, but they comprehend and can communicate the basic principles of geology and supporting sciences. They possess the capacity for critical and independent thinking. They are able to write a fundable research proposal, evaluate current relevant literature, carry out the proposed research, and communicate the results of their research to peers at national meetings and to the general public. They work as consultants on engineering and environmental problems, explorers for new minerals and hydrocarbon resources, researchers, teachers, writers, editors, and museum curators.